



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: COMPLIER DESIGN - (24CS2235F)

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TITLE OF THE PROJECT	Smart Source-to-Instruction Compiler	
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Abstract

The **Smart Source-to-Instruction Compiler** is a compiler-based system designed to translate high-level source code into low-level instructions through a structured and intelligent compilation process. The project demonstrates the major phases of compiler design, including **lexical analysis, syntax analysis, semantic analysis, intermediate code generation, code optimization, and target code generation**. The system takes source code as input, identifies tokens and grammatical structures, validates the program according to predefined rules, and generates an intermediate representation before producing corresponding low-level instructions.

The proposed compiler aims to make the compilation process more **transparent, efficient, and easy to understand** by providing meaningful outputs at each compilation phase. It can identify lexical, syntactic, and semantic errors and provide useful feedback to the user. Optimization techniques are applied to improve the generated intermediate and target code by reducing unnecessary operations and improving execution efficiency. The final output is a set of target instructions that represents the behavior of the original source program.

The project provides a practical implementation of fundamental **compiler design concepts** while demonstrating how a source program is progressively transformed from human-readable code into machine-oriented instructions. It can serve as an educational and experimental platform for understanding the internal working of modern compilers and the relationship between different compilation phases.

Signature of the Guide

HOD